

Creating Cloud-Based Solutions for Business Process Automation

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1. Executive Summary

This report presents the internship experience carried out at Finetech, Business and Services Department, under the project titled “*Creating Cloud-Based Solutions for Business Process Automation.*” The primary objective of this internship was to design and implement cloud-based internal applications to automate key business workflows using the low-code platform Google AppSheet. During the internship period, three main applications were developed:

1. **Internal Service Request Management System (ISRMS)**
2. **Letter Management System (LMS)**
3. **Tender Management System (TMS)**

The idea behind these apps was to organize and improve efficiency and transparency of day-to-day operations in organizations. By cutting down on manual paperwork, we were able to improve coordination across different departments and make data much more visible throughout the organization.

These web applications using Google AppSheet platform, which ties directly into Google Workspace services. Even AppSheet is a low-code platform, it still requires a lot of precise work. I had map out logical structures, design custom workflows, and set up expression-based rules just to ensure the automation behaved exactly as intended. Across all three apps, I integrated features like role-based access control, automated bots, document uploads, and multi-step approval workflows and the csv file download functions.

Building this in a real-world environment came with some challenges. User requirements kept shifting on the fly. On top of that precious testing and bug fixing to insure data safety and reliability. Navigating these roadblocks requires a lot of adaptive problem-solving. I worked closely with the development team and my mentors to coordinate testing and find workarounds.

This was a massive shift from typical assignments for the Faculty of Science, where requirements are usually set in stone from day one. Instead, constant stakeholder feedback drove our iterative refinements. It really highlighted how organizational systems must be flexible and evolve dynamically after review.

Ultimately, this experience gave me practical, hands-on exposure to enterprise-level workflow design. The project demonstrates how low-code cloud platforms can be used effectively to build scalable internal solutions that adapt to a company’s changing needs.

Project Scope

The scope of this project included:

- Analysis of existing manual and semi-digital workflows
- Design and development of three cloud-based applications
- Database schema design using Google Sheets
- Implementation of workflow automation and role-based access
- System testing and iterative refinement

2. Introduction

Business Process Automation (BPA) refers to the use of digital technologies to streamline and automate repetitive organizational processes. In modern organizations, many internal operations such as approvals, document handling, and service requests are still partially manual or managed through emails and spreadsheets. These methods often result in delays, miscommunication, and lack of transparency.

With the advancement of cloud computing, organizations are increasingly adopting cloud-based automation solutions. Cloud platforms provide centralized data storage, remote accessibility, scalability, and reduced infrastructure management. Instead of maintaining local servers and software systems, cloud-based applications allow teams across departments to access and manage workflows through web or mobile interfaces.

During my internship at Finetech Consultancy (Pvt) Ltd, within the Business and Services Department as a Web Developer, I was assigned to develop cloud-based applications to automate selected internal processes. The organization identified three major operational areas that required improvement:

- Internal service request handling
- Official letter management
- Tender management and approval workflows

The main objective of this project was to design and implement three cloud-based applications using the Google AppSheet platform in order to improve efficiency, enforce structured workflows, and reduce manual intervention.

Unlike university-based assignments, this project involved real users from multiple departments. As a result, requirements evolved during development, providing practical exposure to real-world software development dynamics.

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2.1 Cloud-Based Business Process Automation

Business Process Automation (BPA) refers to the use of digital technologies to automate repetitive and structured organizational processes. These processes may include approvals, internal service handling, document tracking, procurement workflows, and reporting activities. According to Denodo (n.d.), BPA improves efficiency, reduces operational errors, and enhances scalability by minimizing manual intervention in routine tasks.

With the evolution of cloud computing, many organizations have shifted from traditional on-premises systems to cloud-based automation platforms. Cloud computing provides computing resources over the internet, allowing organizations to access applications, storage, and processing power without maintaining physical infrastructure. IBM (2024) highlights that cloud computing offers flexibility, efficiency, and strategic value by enabling rapid deployment and scalability of applications.

Cloud-based BPA systems operate under service delivery models such as Software as a Service (SaaS) and Platform as a Service (PaaS). These models allow organizations to host workflow applications on shared infrastructure while maintaining centralized control of data and processes. One of the key advantages of cloud-based automation is accessibility. Employees across different locations can access the same centralized database in real time, which improves collaboration and transparency (Kissflow, 2025).

Scalability is another important benefit. Cloud platforms can dynamically allocate resources based on system demand, reducing downtime and improving reliability (IBM, 2024). In addition, cloud-based systems often include automated backup and redundancy mechanisms, improving data security and availability.

Beyond operational efficiency, modern cloud BPA platforms also generate analytical insights. Redwood Software (2024) notes that workflow automation systems provide visibility into process bottlenecks, approval delays, and task durations. Such insights support data-driven decision-making and continuous improvement.

Overall, the literature indicates that cloud-based BPA contributes significantly to digital transformation initiatives by improving process visibility, reducing manual workload, and increasing operational agility.

2.2 Introduction to the Partner Organization

Finetech Consultancy (Pvt) Ltd is a leading technology service provider and a recognized Google Cloud Premier Partner operating in Sri Lanka. The organization specializes in enterprise digital transformation, cloud infrastructure, and business productivity solutions, primarily leveraging Google Workspace and the Google Cloud Platform (GCP).

During my internship period, I was attached to the Business and Services Department as a Web Developer. This department is responsible for optimizing both internal operations and client-facing digital services, making it an ideal environment to implement modern cloud-based business process automation.

2.2 Low-Code Development Platforms

Low-code development platforms have emerged as a practical solution for rapid application development. These platforms reduce the need for traditional programming by providing visual configuration tools, pre-built components, and workflow engines.

Low-code platforms enable organizations to accelerate modernization and improve agility by simplifying application development. However, low-code does not eliminate the need for technical understanding. Logical structuring, workflow planning, and validation rules still require analytical thinking.

Low-code systems are particularly suitable for internal workflow automation because they allow organizations to quickly adapt applications when requirements change. This flexibility becomes valuable in real-world scenarios where business processes evolve over time.

2.3 Google AppSheet Platform



Google AppSheet is a cloud-based low-code platform that enables users to build applications using structured data sources such as Google Sheets and Cloud SQL. It is integrated within the Google Cloud ecosystem and supports deployment across web and mobile platforms.

According to Google Cloud (2021), AppSheet allows users to create applications faster and with lower development overhead compared to traditional coding approaches. Applications built using AppSheet follow a data-centric architecture, where tables define system behavior and interface components.

Key features of AppSheet include:

- Data-driven application design
- Visual interface builder
- Expression-based logic for validation and workflow control
- Automation bots for notifications and approvals
- Role-based access control

Although AppSheet is categorized as a no-code or low-code platform, it still requires logical configuration through expressions similar to spreadsheet formulas. These expressions define conditional visibility, data validation, workflow transitions, and access permissions.

Industry case studies indicate that AppSheet is widely used for internal business applications such as request management, inventory tracking, and approval workflows. The platform allows organizations to build operational tools quickly without extensive backend development.

In the context of this internship project, AppSheet was selected because it supports:

- Rapid development
- Cloud-native deployment
- Role-based workflow automation
- Easy integration with Google Workspace

However, as observed during development, building complex multi-department workflows still requires careful planning and testing despite the low-code nature of the platform.

2.4 Google Sheet



Google Sheets

Google Sheets serves as the primary relational database backend for applications built on Google AppSheet. As a cloud-native application within Google Workspace, it provides real-time synchronization, accessible data storage, and seamless integration with AppSheet's data-driven architecture. Utilizing Google Sheets eliminates the need for complex database server management while ensuring data integrity through structured tables, rows, and relational keys.

2.5 Objectives of the Study

The organization identified three major operational areas that required improvement: internal service request handling, official letter management, and tender management workflows. The main objective of this project was to design and implement three cloud-based applications using the Google AppSheet platform in order to improve efficiency, enforce structured workflows, and reduce manual intervention.

The specific objectives were:

1. To develop an Internal Service Request Management System (ISRMS) for structured internal request submission and approval.
2. To develop a Letter Management System (LMS) for systematic tracking of incoming and outgoing letters.
3. To develop a Tender Management System (TMS) to automate multi-department approval workflows.

3. Methodology

The project followed an iterative development approach. Requirements were initially provided by the placement mentor and the Business Analyst (BA) of the department. No formal stakeholder interviews were conducted. Instead, basic workflow descriptions and expected functionalities were communicated through internal discussions and documentation.

3.1 Requirement Analysis

Initial baseline requirements were gathered through direct consultations with the placement mentor and the department's Business Analyst (BA) rather than formal stakeholder interviews. These initial discussions provided the core workflow descriptions, required data fields, and basic access control rules.

The requirement analysis process was continuous. As preliminary prototypes were built, they were presented to department users. Cross-departmental feedback was analyzed to identify new system constraints, such as additional approval hierarchies, departmental structuring, and specific data capture needs (e.g., file attachments and CSV exports). This iterative cycle of gathering feedback and integrating it into the system design ensured the final applications accurately mapped to real-world operational dynamics.

3.2 Data Design

All three applications were developed using Google Sheets as the backend relational database. The data design methodology focused on creating a normalized structure to eliminate data redundancy and ensure system scalability.

Each application followed a structured data model comprising:

- **Master Tables:** To store static or rarely changed data (e.g., user roles, department lists).
- **Transactional Tables:** To store dynamic, continuously updated records (e.g., tender requests, logged letters).

These tables were connected using reference columns and unique primary keys within AppSheet to establish referential relationships. The database design was strictly configured to ensure data integrity, control workflow transitions securely, and enforce role-based access management at the data level.

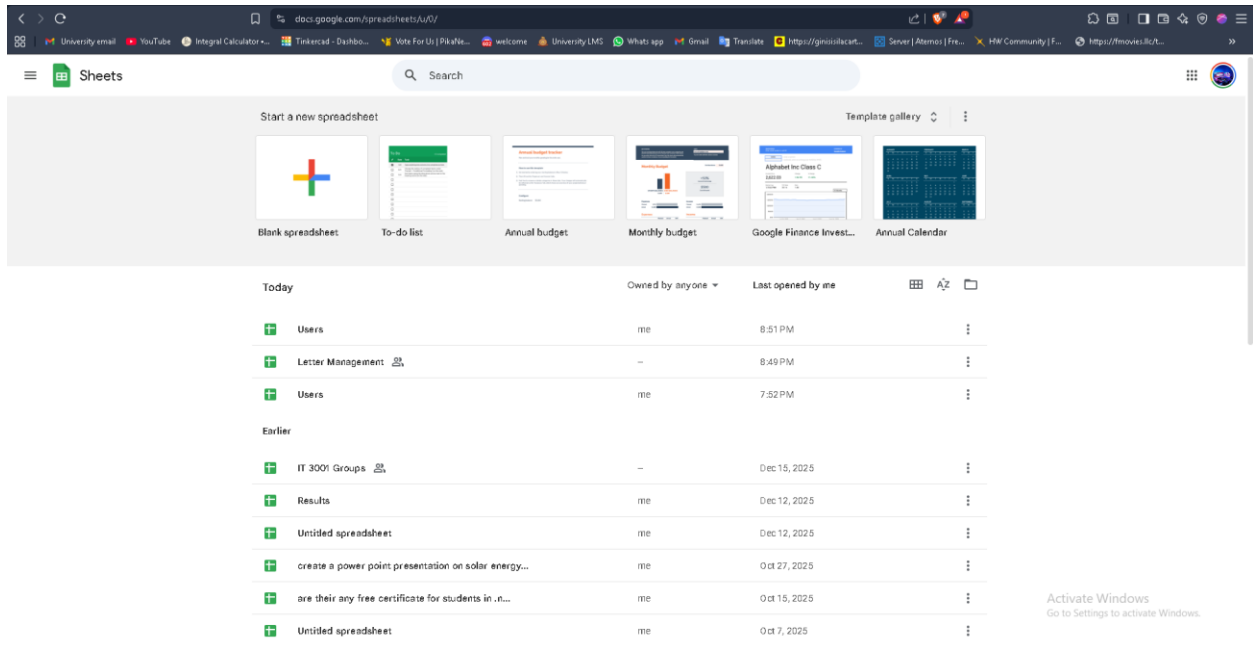


Figure 1 - Google sheets platform

Letter Management

File Edit View Insert Format Data Tools Gemini Extensions Help

Search Menu

100% 123 Roboto 10 B I U A

C17

Summarize this data

	A	B	C	D	E	F	G	H	I	J	K
1	ID	Name	Email	Department	UserRole	JobRole	LetterReceptionView	DepartmentView	OutgoingLettersView	ConfigurationsView	
2	5a2b3c4f	David Lee	david.lee@corp.com	1	Normal user	DR-011					
3	1c9d8e7b	Sarah Chen	sarah.c@corp.com	2	DG	DR-014					
4	6f5g4h3i	Alex Rivera	alex.r@corp.com	4	Admin	DR-020					
5	2j1k0l9m	Hannah Kim	hannah.k@corp.com	3	Chairman	DR-017					
6	7n8o9p0q	John Williams	john.w@corp.com	1	Normal user	DR-009					
7	8r4s5t6u	Priya Sharma	priya.s@corp.com	5	Admin						
8	9v7w6x5y	Omar Hassan	omar.h@corp.com	2	Normal user	DR-012					
9	4z3a2b1c	Chloe Davis	chloe.d@corp.com	3	DG	DR-015					
10	9d0e1f2g	Ben Miller	ben.m@corp.com	5	Normal user						
11	e5f6g7h8	Lisa Wong	lisa.w@corp.com	4	Admin	DR-018					
12	0i9j8k7l	Sam Jones	sam.jones@corp.com	1	Normal user	DR-009					
13	56f1b7e0	Chamindu	chamindu@demo.fcpl.biz	3	Admin	DR-016					
14	2c66tee2	Auravexonflow@gmail.cc		3	Chairman	DR-016					
15	cbcdaf18	ravi	wickramasingha2002@g	5	Chairman						
16											
17											
18											
19											
20											
21											
22											

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Users

Department Roles

Departments

Letters Incoming

Letter Logs

Letters Outgoing

Categories

Attachments

Outgoing Attachments

Comments

Figure 2 - Table structure of letters app

3.3 Application Development Using AppSheet

Google AppSheet was utilized as the primary low-code development platform. Because the core platform and data design principles were identical across all three systems, a common development lifecycle was applied:

1. **Data Integration:** Linking the structured Google Sheets to the AppSheet environment and defining column types (e.g., Enum, File, Ref).
2. **UX/UI Configuration:** Creating interactive interface components, including structured forms for data entry and dashboard views for management oversight.
3. **Behavior Implementation:** Writing AppSheet expressions (similar to spreadsheet formulas) to enforce data validation, conditional field visibility, and workflow control.
4. **Automation Setup:** Configuring automation bots to trigger background actions, such as generating automated email notifications and updating status columns based on user inputs.

3.4 Workflow Implementation

While the underlying development process was the same, the business logic was customized to map the specific workflows of each system:

- **Internal Service Request Management System (ISRMS):**

The implementation logic relied heavily on conditional expressions to dynamically display specific form fields based on request priority. Automation bots were configured to route approval requests to the correct managerial hierarchy based on the requester's assigned department.

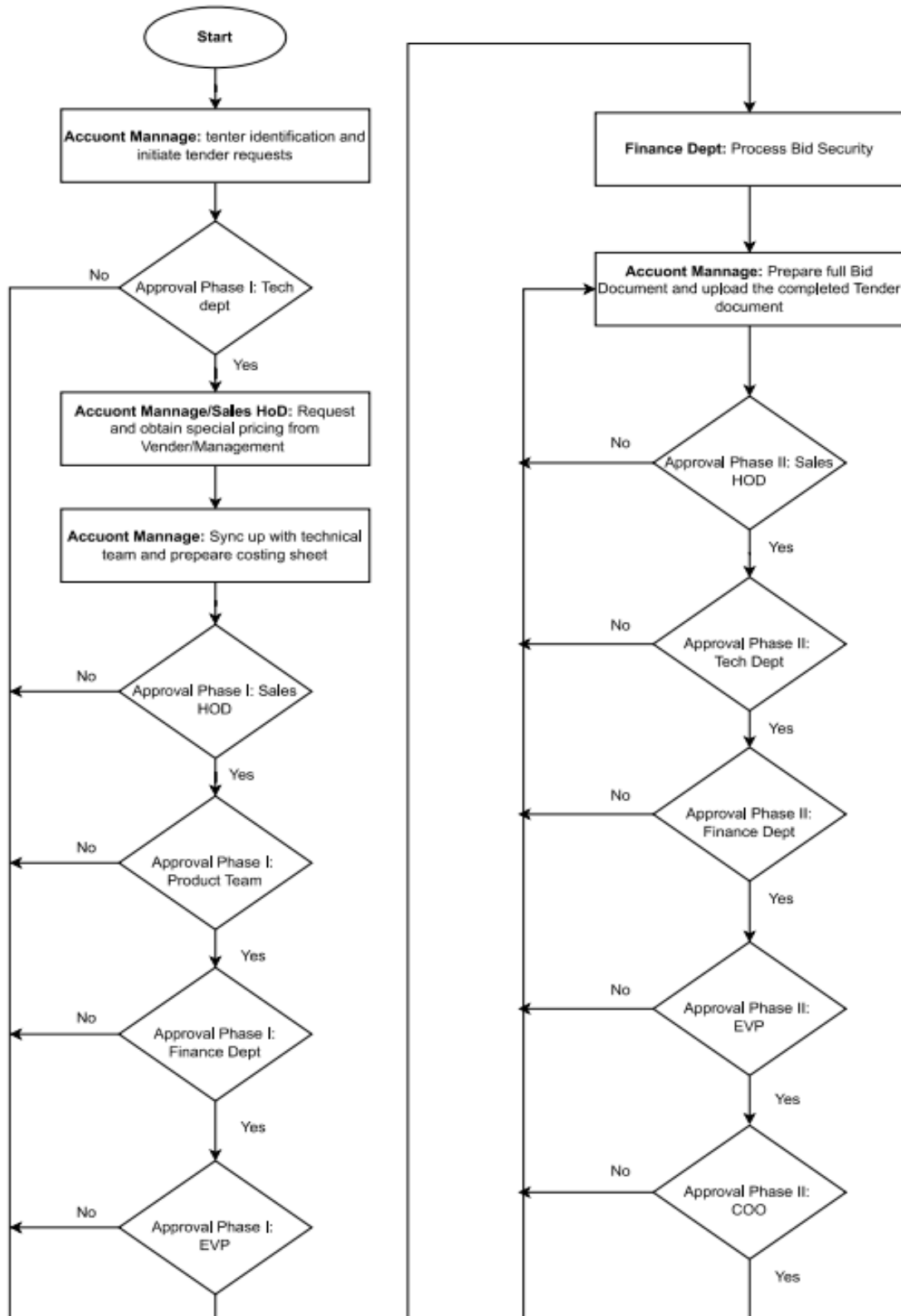
- **Letter Management System (LMS):**

Implementation focused on utilizing action buttons for users to quickly categorize incoming versus outgoing mail. Reference columns were utilized to map scanned documents to specific departments, triggering automated database logging to replace the manual paper registry.

- **Tender Management System (TMS):**

This workflow utilized multi-stage sequential approval chains. The logic was implemented using strict status transitions (e.g., Technical specs → EVP Approved → Finance Validated). Scheduled reminders and automated state changes were configured to handle the complex, multi-department routing.

App Workflow Tender Management System (TMS):



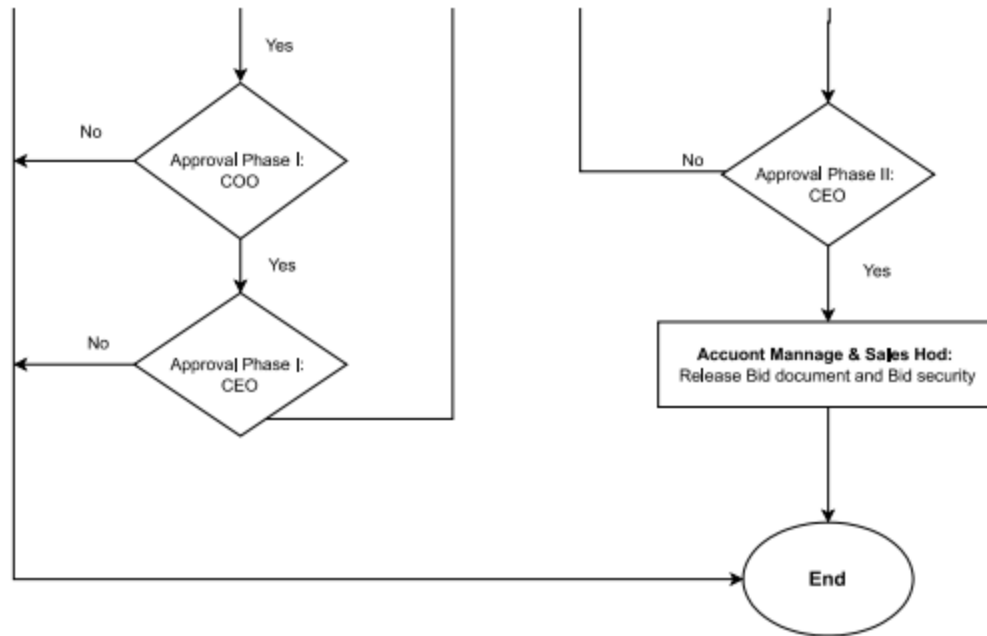


Figure 3 - App Workflow of Tender app

During development of the Tender system, workflow logic required multiple adjustments due to the addition of new departments and approval stages.

For example:

- The EVP role was inserted into the middle of the approval chain.
- Sales departments were separated into three distinct approval paths.
- Finance document validation was added after bid security confirmation.

Each change required updating status transitions and re-testing the entire workflow to ensure no logical conflicts occurred.

App Workflow for Letter Management System (LMS):

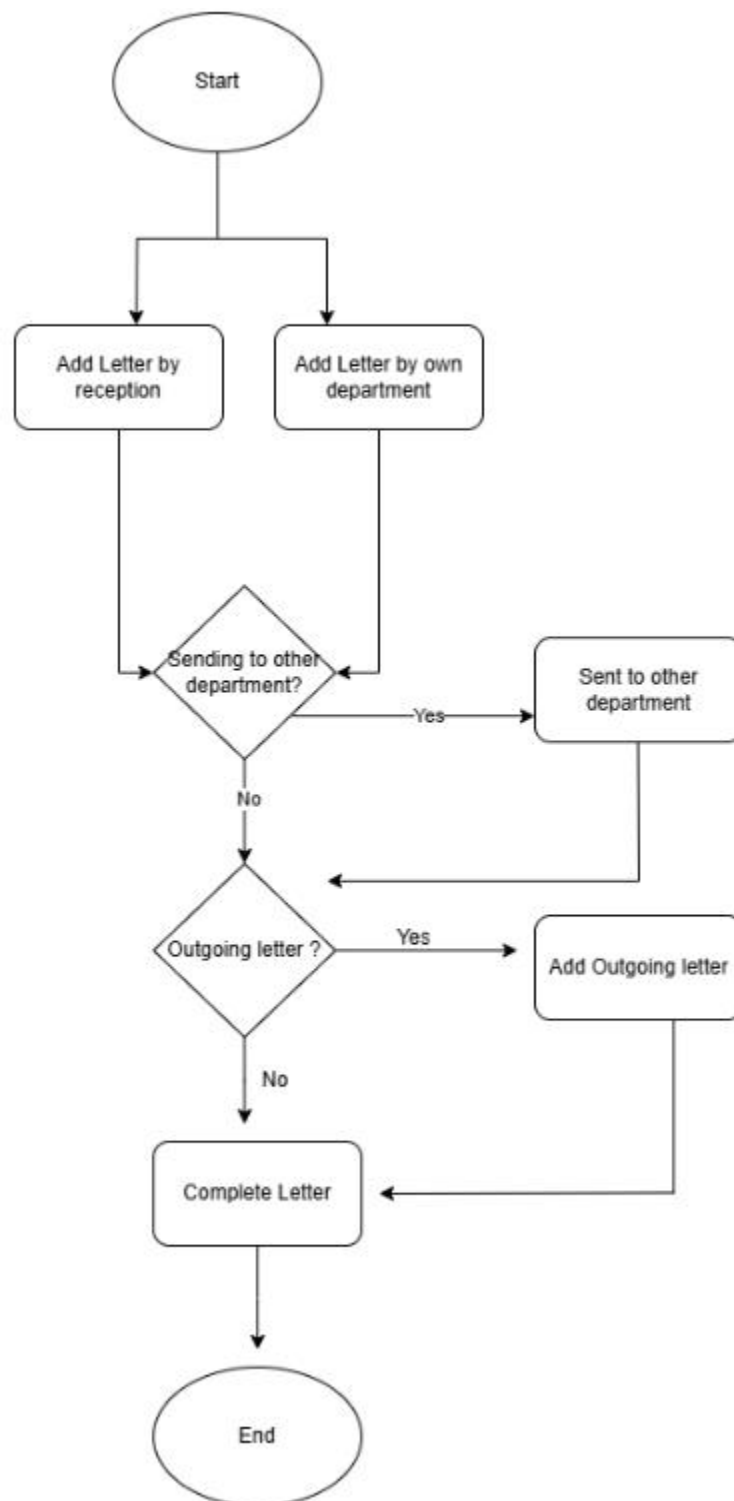


Figure 4 - App workflow of letters app

App workflow for Internal Service Request Management System (ISRMS)

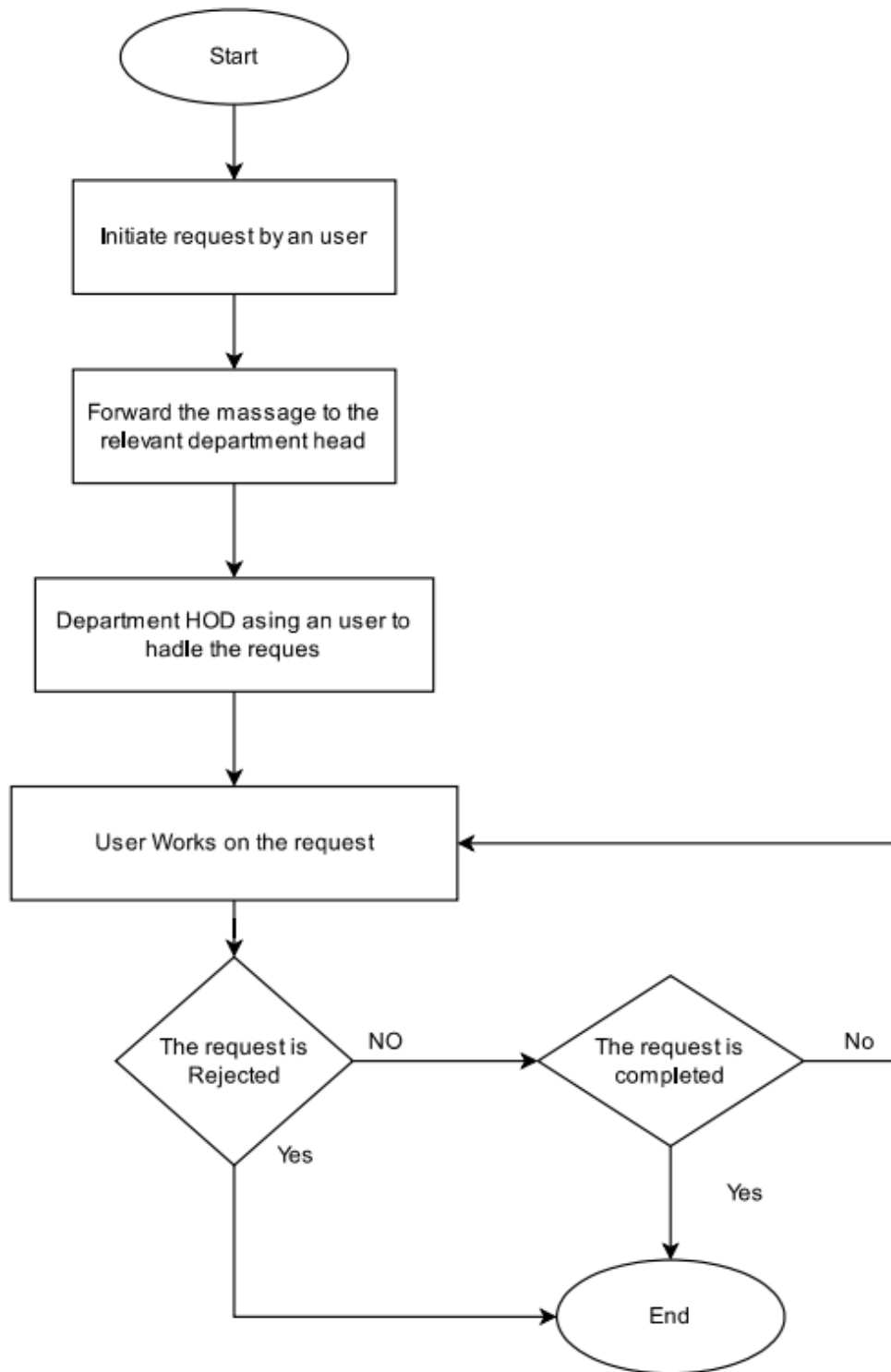


Figure 5 - App workflow for request app

3.5 Testing and Debugging

A standardized testing and validation protocol was applied across all three applications before deployment. The testing methodology included:

- **Simulation Testing:** Creating dummy records and simulating multi-role approval chains using AppSheet's "Preview as user" feature to verify role-based access controls.
- **Cross-Platform Validation:** Testing the applications in both desktop web browsers and mobile preview modes to ensure UI responsiveness and identify platform-specific sync behaviors.
- **Trigger Verification:** Actively testing conditional logic and verifying that email notifications and automation bots fired correctly at the designated workflow stages.
- **Data Consistency Checks:** Monitoring the backend Google Sheets during test inputs to ensure data was writing, updating, and referencing correctly without duplication or data loss.

Final validations were conducted via live demonstrations for the placement mentor and Business Analyst, allowing for final adjustments before the systems were deployed for departmental use.

4. Results and Discussion

4.1 Overall Project Outcomes

By the end of the internship, I successfully deployed three cloud-based applications for Finetech's consultancies and Services Department. Each system was built to solve a specific operational problem by replacing older, manual, or email-heavy processes. The final results of these projects can be broken down into three main areas: better daily operations, smoother workflow automation, and a positive impact on the organization's overall efficiency.

4.2 System Design and Implementation

4.2.1 Internal Service Request Management System (ISRMS)

The first objective was to build the ISRMS app, which lets employees create and track internal service tasks like IT support or maintenance. It is a straightforward application designed to organize staff requests. Instead of relying on random emails or direct messages, the app keeps everything in one manageable place. It also adds transparency, as administrators can see exactly who made a request and when it happened.

When users open the app on their phone or web browser, they are greeted by a "Dashboard" view that summarizes their current requests by status. From there, they can tap "Submit Request" to fill out a form that includes fields for Request Type, Description, Priority, and Requester Name. I made several fields mandatory to ensure no data is left out. (The screenshot below shows the ISRMS home screen.)

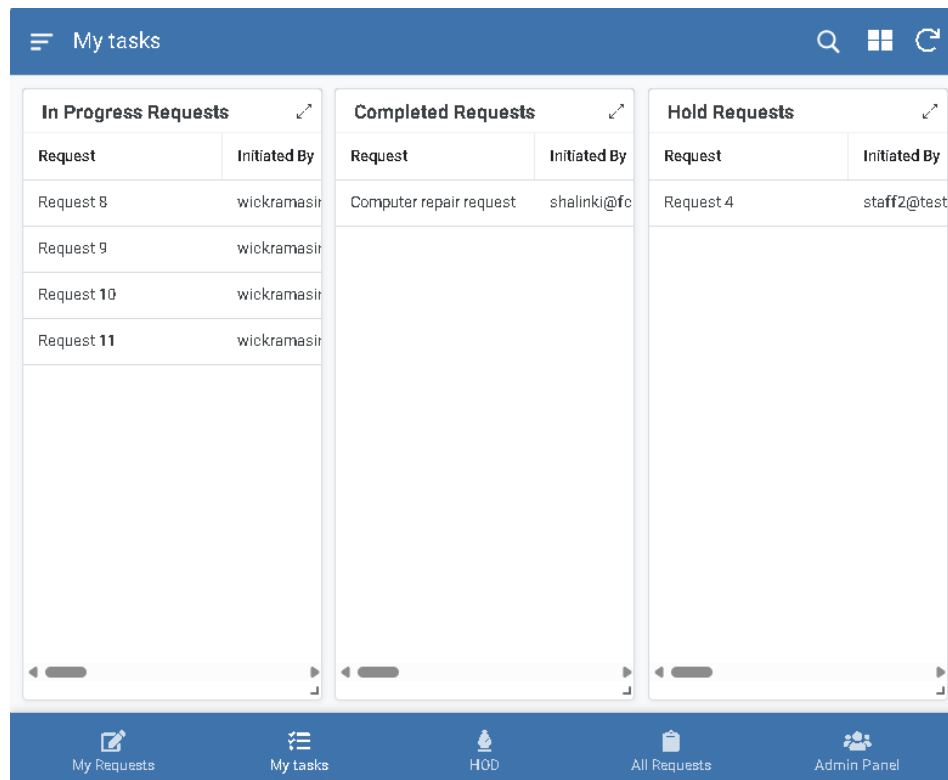


Figure 6 - Request app My task view

- **Key Features:**
- **Request Submission Form:** I created a simple, user-friendly form for logging new requests. Using AppSheet expressions, I added conditional logic so that if an employee selects “High Priority,” a new “Justification” field automatically pops up.

The screenshot shows a mobile app interface for submitting requests. The form is titled "My Requests Form" and includes the following fields and controls:

- Initiated By*:** A text input field containing the email address "wickramasingha2002@gmail.com".
- Request Date Time*:** A date and time picker showing "03/07/2026 03:43:24 PM" with a calendar icon on the right.
- Request:** A large text area for entering the details of the request.
- Category:** Three buttons labeled "Category 1", "Category 2", and "Category 3" for selecting the request category.
- Assigned Department:** A dropdown menu for selecting the department to which the request is assigned.
- Buttons:** "Cancel" and "Save" buttons at the bottom of the form.

Figure 7 - Form for adding a new request

- **HOD Approval Workflow:** Once a request is submitted, the system automatically routes it to the requester’s Head of Department (HOD). HODs have a specific “Pending Approvals” view where they can see incoming requests, review the details, and assign them to the right staff members. This action automatically updates the status of the request.

Requests to Assign

Request	Initiated By
Request 12	shalinki@test.com

Assigned Requests

Request	Initiated By
In Progress	
Request 7	wickramasingha2002@gmail.com
Request 11	wickramasingha2002@gmail.com
request Test	wickramasingha2002@gmail.com
Test Request	shalinki@test.com
Completed	
Computer repair request	shalinki@test.com
Approval for a letter	shalinki@test.com

HOD To Assign form

Request Date Time*
01/21/2026 02:45:57 PM

Request
Request 12

Category
Category 1

Assigned Department
Department 1

Assigned User*

Cancel Save

Figure 8 - HOD view and form for assign an user for the task

- **Status Tracking:** The app gives both regular employees and managers a live list of tasks. Employees can track their own submissions through labels like In-Progress, Hold, or Completed, while managers have access to active tasks and a full archive of past requests.

Admin Panel

User Management

Name ↑	id	UserEmail
	926c6550	staff3@test.com
	926c6551	staff4@test.com
Bob	3	hod1@test.com
Chamindu	2	wickramasingha2002@gmail.com
Chamindu	926c6549	staff2@test.com
Charlie	4	hod2@test.com
David	5	hod3@test.com
Eve	6	hod4@test.com
Frank	7	staff1@test.com
Nishanthan	2cbb369	staff3@test.com
Dehudu	778c3e18	staff4@test.com

Department Management

Name	id	UserEmail
Chamindu	2	wickramasingha2002@gmail.com
User	fc291845	user@example.com
Bob	3	hod1@test.com
Charlie	4	hod2@test.com
David	5	hod3@test.com
Eve	6	hod4@test.com
Frank	7	staff1@test.com
Chamindu	926c6549	staff2@test.com
	926c6550	staff3@test.com
	926c6551	staff4@test.com
Shalinki	18	shalinki@test.com

My Requests My tasks HOD All Requests Admin Panel

Figure 9 - Admin panel of the app

Overall, this design follows standard business process automation practices. By letting the app handle routing and status updates, we removed the need for manual handoffs like paper forms or back-and-forth emails. It also creates a solid audit trail because every action is timestamped and tied to a specific user, making the whole process much more reliable.

Operational Improvements

Before this project, staff handled internal requests using informal channels like emails or direct messages. This old method caused delays, made it hard to track progress, and gave managers poor visibility over what was getting done.

After rolling out the app, the workflow completely changed:

- Staff now submit requests through proper, structured forms.
- The system assigns a unique tracking ID to every single request.
- Department heads have a dedicated space to view pending and approved tasks.
- Everyone involved can see real-time status updates.
- I set up role-based security, ensuring people only see what they are authorized to view.

Measurable Improvements

By achieving this objective, the department saw some immediate, practical benefits:

- A major drop in the time wasted on manual follow-ups.
- Much better transparency across the board for tracking tasks.
- A single, centralized database that securely stores all past records.

4.2.2 Letter Management System (LMS)

The Letter Management System was designed to digitize the logging of official incoming and outgoing mail. Because this was the very first application I developed during my internship, it served as my foundational learning experience with the AppSheet platform.

When administrators or staff open the app, they are presented with a central Dashboard. I utilized AppSheet's built-in charting tools to display a summary chart on this dashboard, which gives management a quick visual breakdown of letter volume trends by type and month.

The implemented features that fulfill the system's requirements include:

- **Dashboard View:** Administrators have access to a summary chart (built-in AppSheet chart) showing the number of letters by type or month. This provides quick insight into letter volume trends. (Screenshot below shows the LMS dashboard listing logged letters.)

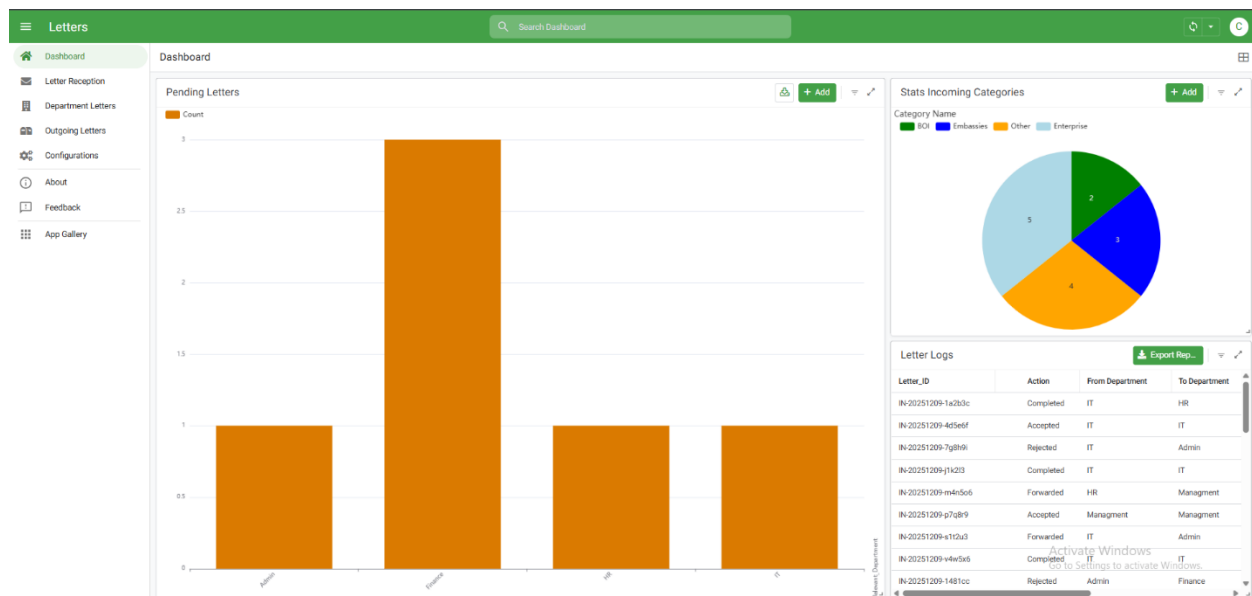


Figure 10 - Letters app Dashboard view

- **Key Features:**
- **Letter Log Form:**

A straightforward data entry form where users simply select "Incoming" or "Outgoing." They then fill in standard fields like the Date, Sender/Recipient, Subject, and Reference Number. I also added a file upload option so staff can directly attach a scanned copy of the physical document.

Letter Reception

New Letters

Ref_No	Status	Category	Relevant_Department
IN-20251209-7y8r9i	Pending	Embassies	Finance
IN-20251209-812u3	Pending	HR	HR
IN-20251209-81450d8	Pending	Other	Finance
IN-20251207-C33a4f5d	Pending	Enterprise	Admin
IN-20251209-9icf7	Pending	Embassies	Finance
IN-20251210-8a8t	Pending	BOI	IT

Returned Letters

Ref_No	Sender	Received_Date
IN-20251209-v6w5a6	SarahDoe@acdf.it	12/5/2025

Completed Letters

Ref_No	Sender	Received_Date
IN-20251209-1a2b3c	AcmeCorp@gmail.com	12/5/2025
IN-20251209-1jk2l3	local.council@example.com	12/5/2025
IN-20251209-m4n5o6	marketing.agency@example.com	12/6/2025
IN-20251209-p7q8r9	customerservice@example.com	12/5/2025
IN-20251209-1481cc	test.abcd@demo.fopl.biz	12/5/2025

Figure 11 - Letters app incoming letters view

Outgoing Letters

Outgoing Pending

Linked_Letter_ID	Subject	Receiver	Send_Date	Delivery_Method
IN-20251209-m4n5o6	Preliminary Tender Submission	Ministry of Works	12/3/2025	Air Mail
IN-20251209-1481cc	Annual Audit Request Documents	External Auditors LLP	12/2/2025	Courier

Outgoing Completed

Outgoing_Ref_No	Linked_Letter_ID	Status	Subject	Receiver
OUT-20251209-1j4k5	IN-20251209-p7q8r9	Completed	Follow-up on Payment Invoice 456	Customer Z Corp

Figure 12 - Letters app outgoing letters view

- **Approval/Review Workflow:** For incoming letters that require review, we added an optional step where a manager can mark a letter as “Reviewed” and add notes. This was done by adding a workflow button that changes a “Reviewed” column from No to Yes. If integrated with email, the system could notify concerned staff (not implemented in this phase).
- **Searchable List:** The main view lists all letters in a table with key fields (Date, Type, Subject). Users can filter by date range or search by subject/party.
- **Notifications:** Though not critical for every use-case, we configured an automation so that when a new incoming letter is logged, a notification can be sent to a mailing list (configurable via a Google Group).

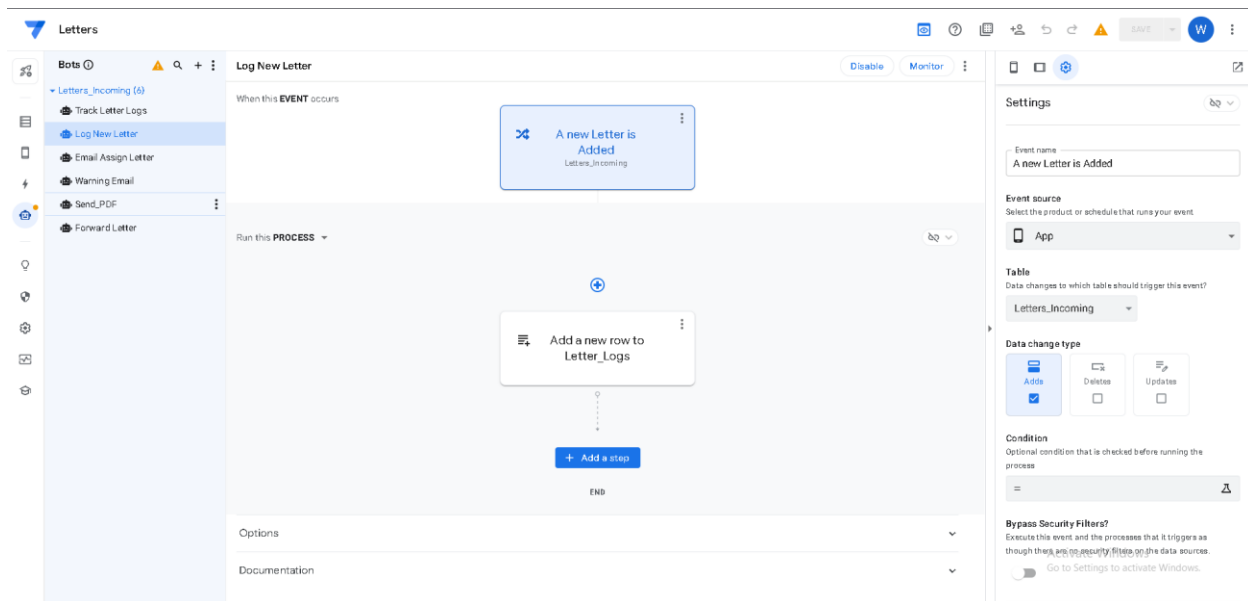


Figure 13 - Automation bots for the letters app

By replacing spreadsheets or paper logs, the LMS ensures that every letter's details are consistently captured. The app enforces structured data entry and makes it easy to retrieve records. In practice, department staff reported that entering a letter into the app took just a few taps on their phone, much simpler than filing a physical log.

The Letter Management System was the first application developed during the internship and served as a foundational learning experience.

Functional Outcomes

- Registration of incoming and outgoing letters.
- Digital record maintenance replacing manual logbooks.
- Search and filter features for quick retrieval.
- CSV export functionality for reporting.
- Role-based access for administrative users.

Development Challenges

Since this was the first AppSheet project undertaken, several challenges were encountered like understanding expression syntax for computed columns and implementing conditional actions and configuring automation bots independently.

Support from fellow developers helped resolve automation logic issues and optimize workflow triggers.

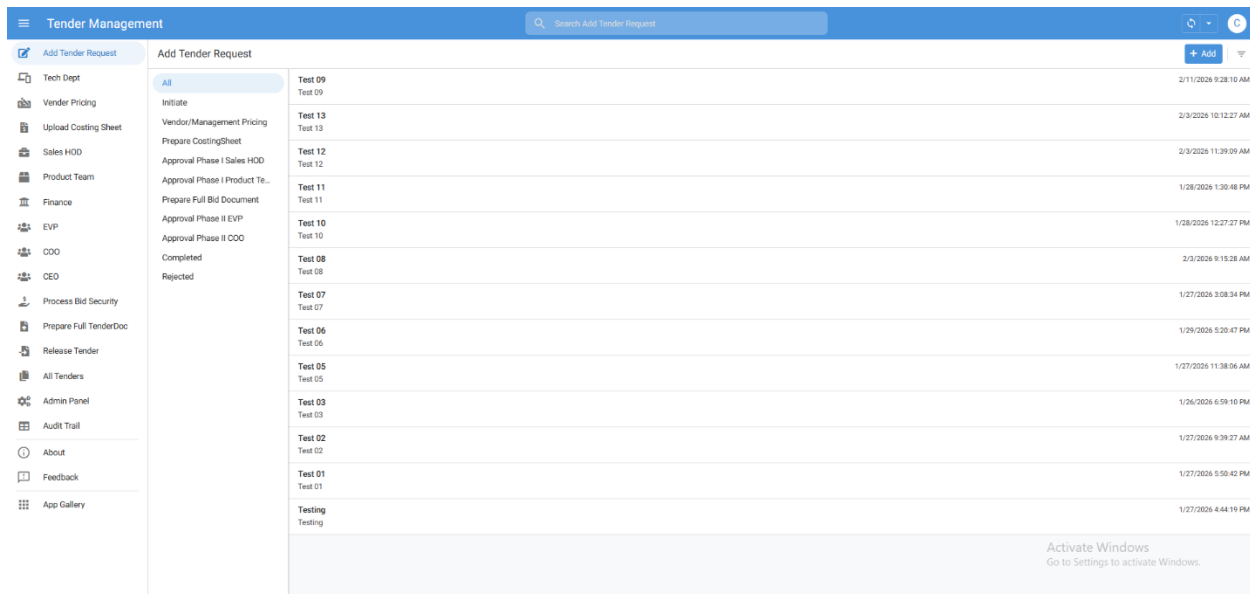
Organizational Impact

This project improved document traceability, reduced dependency on physical registers.

4.2.3 Tender Management System (TMS)

The Tender Management System was the primary focus of my internship and easily the most complex of the three applications I built. It is designed to support Finetech's procurement process by tracking tenders and bids from start to finish. Because procurement involves a lot of oversight, the workflow I designed contains 21 distinct steps, requiring a dual-approval process from each participating department.

When users log into the app, they see views for both "Active Tenders" and "Closed Tenders." Each tender entry holds critical data like the Tender Title, Budget, Deadline, and current Status.



The screenshot displays the 'Tender Management' app interface. On the left is a sidebar menu with options like 'Add Tender Request', 'Tech Dept', 'Vendor Pricing', 'Upload Costing Sheet', 'Sales HOD', 'Product Team', 'Finance', 'EVP', 'COO', 'CEO', 'Process Bid Security', 'Prepare Full TenderDoc', 'Release Tender', 'All Tenders', 'Admin Panel', 'Audit Trail', 'About', 'Feedback', and 'App Gallery'. The main area shows a table of tender entries. The table has columns for 'Status', 'Tender Title', and 'Date'. The entries are as follows:

Status	Tender Title	Date
Test 09	Test 09	2/11/2026 9:28:10 AM
Test 13	Test 13	2/3/2026 10:12:27 AM
Test 12	Test 12	2/3/2026 11:39:09 AM
Test 11	Test 11	1/28/2026 1:30:48 PM
Test 10	Test 10	1/28/2026 12:27:27 PM
Test 08	Test 08	2/3/2026 9:15:26 AM
Test 07	Test 07	1/27/2026 3:08:34 PM
Test 06	Test 06	1/28/2026 5:20:47 PM
Test 05	Test 05	1/27/2026 11:38:06 AM
Test 03	Test 03	1/26/2026 6:59:10 PM
Test 02	Test 02	1/27/2026 9:39:27 AM
Test 01	Test 01	1/27/2026 5:50:42 PM
Testing	Testing	1/27/2026 4:44:19 PM

Figure 14 - Account manager view (for add new tenders and track the started ones)

The key features I implemented to meet the department's requirements include:

- **Key Features:**
- **Tender Entry Form:** A form to create or edit a tender record. Required fields include *Title*, *Project Owner*, *Budget*, and *Deadline*. The form also allows attaching scope documents.
- **Bid Submission Tracking:** For each tender, the app has a related “Bids” table where individual vendor bid details can be entered. Users enter the bidder name, bid amount, and date. This was implemented using a “Ref” column in AppSheet linking Bids to the parent Tender record.
- **Approval Workflow:** Once bidding is complete, the procurement manager can select a winning bid. We implemented this by adding an action button “Select Winner” on the tender detail view. When pressed, the tender status changes to “Awarded” and the selected bid is flagged. An email notification is sent to inform the project owner of the decision.

- **Timeline and Notifications:** We set up an automated reminder: if the tender's deadline column passes without status change, the app (via a scheduled bot) sends a reminder to the procurement team to take action.

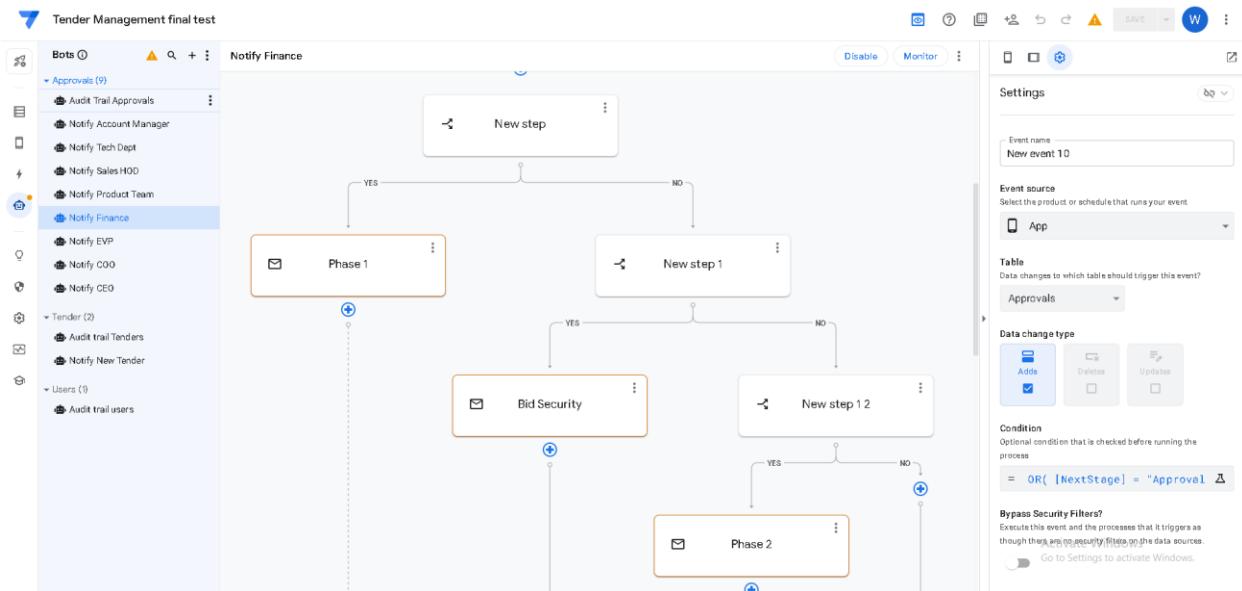


Figure 15 - Automation configurations for tenders app

- **Reporting:** A summary dashboard shows active tenders with upcoming deadlines and closed tenders by award date. This helps stakeholders stay on top of procurement timelines.

The TMS app replaces a fragmented process of paper proposals and email chains. All tender data is now centralized. For example, when multiple bids came in for the last tender we tested, they were entered directly into the app; the manager could then compare them side-by-side on a tablet and mark the winner with one tap. This streamlined the award process and maintained a clear record of all bids

The Tender Management System was the most complex of the three applications, involving multiple departments and approval stages.

Initial Workflow Design

The original workflow consisted of:

- Submission
- Department review
- Approval stages
- Final decision

However, after the first testing phase, major changes were requested.

4.2.3.1 Workflow Modification Challenges

Unlike academic projects where requirements are fixed and the scope is strictly controlled, this industry project proved to be highly dynamic. After we completed the first testing phase, the stakeholders requested major changes to the workflow.

For example, I had to insert a brand new Executive Vice President (EVP) approval stage right into the middle of the existing flow. Additionally, the Sales Department was restructured into three distinct sub-departments, and the Finance team requested a new feature to attach documents *after* they marked a bid security as processed. Mid-development, they also asked for the ability to upload images directly from mobile devices, rather than just linking files.

Implementing these changes meant I had to completely update the workflow logic, add new database tables, adjust my reference relationships, and reconfigure the automation bots. It was a challenging but highly practical lesson in how real-world user requirements evolve once people actually start interacting with a system.

These modifications required:

- Updating workflow logic.
- Adding new database tables.
- Adjusting reference relationships.
- Reconfiguring automation bots.
- Modifying role-based conditions.

4.2.3.2 Technical Issues Faced

One significant issue occurred in child tables:

- Duplicate entries were allowed.
- Records occasionally failed to save in desktop view.
- The issue did not appear in mobile view.

Support was requested from Google AppSheet support. Later, the issue was resolved with assistance from two fellow developers by restructuring reference keys and validation logic.

A three-way data validation is implement to solve this problem. Approvals table [PreviousStage] formula is written to get the most recent approval prom the table

Expression Assistant



App Formula for column PreviousStage (Ref)



```

1  MAXROW(
2    "Approvals",
3    "Timestamp",
4    AND(
5      [Timestamp] < [_THISROW].[Timestamp],
6      [TenderID] = [_THISROW].[TenderID]
7    )
8  )

```



Test

Figure 16 - function to get the previous row of approvals table

Then the [currentstage] formula is implemented to get the data from “previousstage row” “nextstage column”

Expression Assistant



Valid If formula for column CurrentStage (Yes/No,List)



IFS(condition1, value1, [condition2, value2 ...])

```

1  IFS(
2    OR(
3      ISBLANK([PreviousStage]),
4      [PreviousStage].[NextStage] = "Initiate"
5    ),
6    [_THIS] = "Initiate",
7
8    [PreviousStage].[NextStage] = "Approval Phase I Tech",
9    [_THIS] = "Approval Phase I Tech",
10
11   [PreviousStage].[NextStage] = "Prepare CostingSheet",
12   [_THIS] = "Prepare CostingSheet",
13
14   [PreviousStage].[NextStage] = "Vendor/Management Pricing",
15   [_THIS] = "Vendor/Management Pricing",
16
17   [PreviousStage].[NextStage] = "Approval Phase I Sales HOD",
18   [_THIS] = "Approval Phase I Sales HOD",
19
20   [PreviousStage].[NextStage] = "Approval Phase I Product Team",
21   [_THIS] = "Approval Phase I Product Team",
22

```



Test

Examples

Data Explorer



[Help me with expressions](#)

Cancel

Save

Figure 17 - code for currentstage data validation

And then the [nextstage] column ,to calculate the next stage of the flow if follows the dessition selected by the user and the login is restricted by the appsheet form views to insuere it work flow go smoothly.

```

IFS(
  CONTEXT("View") = "InitiateTenders_Form",
  LIST("Approval Phase I Tech"),

  CONTEXT("View") = "VendorMangementPricing_Form",
  IFS(
    [Decision] = "Send Back", LIST("Approval Phase I Tech", "Initiate"),
    TRUE, LIST("Prepare CostingSheet")
  ),

  CONTEXT("View") = "uploadCostingSheet_Form",
  IFS(
    [Decision] = "Send Back", LIST("Vendor/Management Pricing", "Approval Phase I Tech", "Initiate"),
    TRUE, LIST("Approval Phase I Sales HOD")
  ),

  CONTEXT("View") = "TechDeptView_Form",
  IFS(
    [Decision] = "Reject", LIST("Rejected"),
    [Decision] = "Send Back", LIST("Initiate"),
    TRUE, LIST("Vendor/Management Pricing")
  ),

  CONTEXT("View") = "SalesHODView_Form",
  IFS(
    [Decision] = "Reject", LIST("Rejected"),
    [Decision] = "Send Back", LIST("Prepare CostingSheet", "Vendor/Management Pricing", "Approval Phase I Tech", "Initiate"),
    TRUE, LIST("Approval Phase I Product Team")
  ),

  CONTEXT("View") = "ProductDeptView_Form",
  IFS(
    [Decision] = "Reject", LIST("Rejected"),
    [Decision] = "Send Back", LIST("Approval Phase I Sales HOD", "Prepare CostingSheet", "Vendor/Management Pricing", "Approval Phase I Tech", "Initiate"),
    TRUE, LIST("Approval Phase I Finance")
  ),

  CONTEXT("View") = "FinanceDeptView_Form",
  IFS(
    [Decision] = "Reject", LIST("Rejected"),
    [Decision] = "Send Back", LIST("Approval Phase I Product Team", "Approval Phase I Sales HOD", "Prepare CostingSheet", "Vendor/Management Pricing", "Approval Phase I Tech", "Initiate"),
    TRUE, LIST("Approval Phase I EVP")
  ),

  CONTEXT("View") = "EVPView_Form",
  IFS(
    [Decision] = "Reject", LIST("Rejected"),
    [Decision] = "Send Back", LIST("Approval Phase I Finance", "Approval Phase I Product Team", "Approval Phase I Sales HOD", "Prepare CostingSheet", "Vendor/Management Pricing", "Approval Phase I Tech", "Initiate"),
    TRUE, LIST("Approval Phase I COO")
  ),

  CONTEXT("View") = "COOView_Form",
  IFS(
    [Decision] = "Reject", LIST("Rejected"),
    [Decision] = "Send Back", LIST("Approval Phase I EVP", "Approval Phase I Finance", "Approval Phase I Product Team", "Approval Phase I Sales HOD", "Prepare CostingSheet", "Vendor/Management Pricing", "Approval Phase I Tech", "Initiate"),
    TRUE, LIST("Approval Phase I CEO")
  ),

  CONTEXT("View") = "CEOView_Form",
  IFS(
    [Decision] = "Reject", LIST("Rejected"),
    [Decision] = "Send Back", LIST("Approval Phase I COO", "Approval Phase I EVP", "Approval Phase I Finance", "Approval Phase I Product Team", "Approval Phase I Sales HOD", "Prepare CostingSheet", "Vendor/Management Pricing", "Approval Phase I Tech", "Initiate"),
    TRUE, LIST("Bid Security Processing")
  ),

  CONTEXT("View") = "TechDeptView 2_Form",
  IFS(
    [Decision] = "Send Back", LIST("Approval Phase II Sales HOD", "Prepare Full Bid Document"),
    TRUE, LIST("Approval Phase II Finance")
  ),

  CONTEXT("View") = "SalesHODView 2_Form",
  IFS(
    [Decision] = "Send Back", LIST("Prepare Full Bid Document"),
    TRUE, LIST("Approval Phase II Tech")
  ),

  CONTEXT("View") = "FinanceDeptView 2_Form",
  IFS(
    [Decision] = "Send Back", LIST("Approval Phase II Tech", "Prepare Full Bid Document"),
    TRUE, LIST("Approval Phase II EVP")
  ),

  CONTEXT("View") = "EVPView 2_Form",
  IFS(
    [Decision] = "Send Back", LIST("Approval Phase II Finance", "Prepare Full Bid Document"),
    TRUE, LIST("Approval Phase II COO")
  ),

  CONTEXT("View") = "COOView 2_Form",
  IFS(
    [Decision] = "Send Back", LIST("Approval Phase II EVP", "Prepare Full Bid Document"),
    TRUE, LIST("Approval Phase II CEO")
  ),

  CONTEXT("View") = "CEOView 2_Form",
  IFS(
    [Decision] = "Reject", LIST("Rejected"),
    [Decision] = "Send Back", LIST("Approval Phase II COO", "Prepare Full Bid Document"),
    TRUE, LIST("To Release Bid Doc")
  ),

  CONTEXT("View") = "Process Bid Security_Form",
  IFS(
    [Decision] = "Send Back", LIST("Approval Phase I CEO"),
    TRUE, LIST("Prepare Full Bid Document")
  ),

  CONTEXT("View") = "Prepare Full TenderDoc_Form",
  LIST("Approval Phase II Sales HOD"),

  CONTEXT("View") = "Release Tender_Form",
  IFS(
    [Decision] = "Send Back", LIST("Approval Phase II CEO"),
    TRUE, LIST("Completed")
  )
)

```

Figure 18 - code to calculate next stage base on user decision

4.2.3.3 Feature Expansion Requests

Mid-development feature changes included:

- Allowing image uploads in addition to files and links.
- Adding new attachment handling logic.
- Introducing department-specific workflow rules.

These changes required redesigning parts of the data schema.

4.2.3.4 Overall Impact to Organization

The developed applications contributed to:

- Process transparency.
- Digital record maintenance.
- Reduced manual paperwork.
- Improved approval monitoring.
- Centralized cloud-based data storage.

The organization now possesses scalable cloud systems that can be further expanded in the future.

4.3 Testing and Validation

For each application, we executed a testing protocol to validate functionality against the stated requirements. Test cases were written for key scenarios (e.g. submit and approve a request, log an outgoing letter and verify it appears in the list, close a tender and check notifications). We performed these tests manually, simulating the different user roles. All core functions were verified to work correctly. Minor issues found (such as a misconfigured email template or a permission oversight) were quickly resolved in the AppSheet Editor and then retested.

AppSheet's real-time preview feature greatly aided testing: any change we made in the editor was immediately visible on the emulator. We also tested the apps on actual devices (Android phone and web browser) to ensure the UI was responsive. In some cases, we tweaked the UX layout (for example, adjusting column widths and grouping fields) to improve readability on mobile screens. By the end of the testing phase, each app operated as designed, and mentors were able to use them without additional training beyond a short introduction.

5. Conclusions

Looking back at the entire internship, the project focused on creating cloud-based solutions for business process automation successfully achieved all of its core objectives. During this three months time period at Finetech, I designed and delivered three custom business applications—the ISRMS, LMS, and TMS using Google AppSheet. I was able to translate the basic requirements provided by my mentor into fully functional workflows much faster than traditional coding would allow because the goole appsheet allow to rapid development of webapps . Through iterative testing and careful configuration, these applications successfully replaced the department's manual, ad-hoc routines with structured, reliable digital solutions.

The development process, however, was not passed without any issues. On the technical side, I had to overcome a steep learning curve regarding AppSheet's specific expression syntax and figure out how to work around some of the platform's user interface and advanced logic limitations and how automations are work. Organizationally, building complex workflows without conducting formal user interviews meant I had to rely heavily on mentor guidance and clarify ambiguous business rules on the fly as user feedback rolled in. The challenges I faced during the development ultimately became the most valuable part of the experience. This significantly improved my practical skills and giving me hands-on experience in business analysis, process mapping, and systematic testing, while also sharpening my soft skills through close collaboration with the Finetech team.

Ultimately, during this internship proved to me just how effectively cloud-based, low-code applications can automate routine business processes without relying on 3 to 6 moths lifecycles of full code base app development. The final results of my work align perfectly with the background literature I reviewed: cloud BPA drastically increases operational efficiency, flexibility and the transparency of work, while low-code platforms enable developers to deliver those solutions rapidly. Overall, the experience highlighted the massive real-world value of combining modern cloud technology with practical, user-focused problem-solving to achieve meaningful improvements in an organization's daily operations.

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